

Week 5-Homework

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load *pima*

```
library(faraway)
data(pima)
```

(a)

```
summary(pima)
```

```
##      pregnant      glucose      diastolic      triceps
## Min.   : 0.000   Min.   : 0.0   Min.   : 0.00   Min.   : 0.00
## 1st Qu.: 1.000   1st Qu.: 99.0   1st Qu.: 62.00   1st Qu.: 0.00
## Median : 3.000   Median :117.0   Median : 72.00   Median :23.00
## Mean   : 3.845   Mean   :120.9   Mean   : 69.11   Mean   :20.54
## 3rd Qu.: 6.000   3rd Qu.:140.2   3rd Qu.: 80.00   3rd Qu.:32.00
## Max.   :17.000   Max.   :199.0   Max.   :122.00   Max.   :99.00
##      insulin      bmi      diabetes      age
## Min.   : 0.0   Min.   : 0.00   Min.   :0.0780   Min.   :21.00
## 1st Qu.: 0.0   1st Qu.:27.30   1st Qu.:0.2437   1st Qu.:24.00
## Median :30.5   Median :32.00   Median :0.3725   Median :29.00
## Mean   :79.8   Mean   :31.99   Mean   :0.4719   Mean   :33.24
## 3rd Qu.:127.2   3rd Qu.:36.60   3rd Qu.:0.6262   3rd Qu.:41.00
## Max.   :846.0   Max.   :67.10   Max.   :2.4200   Max.   :81.00
##      test
## Min.   :0.000
## 1st Qu.:0.000
## Median :0.000
## Mean   :0.349
## 3rd Qu.:1.000
## Max.   :1.000
```

Values of 0 for variables such as glucose, diastolic, triceps, insulin, and BMI are biologically implausible.(It looks like missing values were coded as 0 in this dataset.) Therefore, these zero values were treated as missing data.

```
pima$diastolic[pima$diastolic==0] <- NA
pima$glucose[pima$glucose==0] <- NA
pima$triceps[pima$triceps==0] <- NA
pima$insulin[pima$insulin==0] <- NA
pima$bmi[pima$bmi==0] <- NA
```

(b)

```
r<-glm(test~.,family="binomial",data=pima)
summary(r)
```

```
##
## Call:
## glm(formula = test ~ ., family = "binomial", data = pima)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.004e+01  1.218e+00  -8.246  < 2e-16 ***
## pregnant      8.216e-02  5.543e-02   1.482  0.13825
## glucose       3.827e-02  5.768e-03   6.635  3.24e-11 ***
## diastolic     -1.420e-03  1.183e-02  -0.120  0.90446
## triceps       1.122e-02  1.708e-02   0.657  0.51128
## insulin      -8.253e-04  1.306e-03  -0.632  0.52757
## bmi          7.054e-02  2.734e-02   2.580  0.00989 **
## diabetes      1.141e+00  4.274e-01   2.669  0.00760 **
## age          3.395e-02  1.838e-02   1.847  0.06474 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 498.10  on 391  degrees of freedom
## Residual deviance: 344.02  on 383  degrees of freedom
## (376 observations deleted due to missingness)
## AIC: 362.02
##
## Number of Fisher Scoring iterations: 5
```

The model's residual deviance is 344.02 with 383 degrees of freedom.

The p-value is approximately 0, indicating that at least one predictor is significantly associated with the probability of testing positive for diabetes.

$$T = \text{null deviance} - \text{deviance} = 498.10 - 344.02 \sim \chi^2_{df=391-383}$$

```
T<-r$null.deviance-r$deviance #test statistics
T.df<-r$df.null-r$df.residual
1-pchisq(T,T.df) # p-value
```

```
## [1] 0
```

The p-value ≈ 0 , reject H_0 . This indicates that the predictors provide significant information for predicting diabetes status.

(c)

```
summary(pima$bmi)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##    18.20   27.50   32.30   32.46   36.60   67.10      11
```

```
r$coefficient[7]
```

```
##      bmi
## 0.07053758
```

Woman A: bmi = 1st quantile ≈ 27.5

Woman B: bmi = 3rd quantile ≈ 36.6

The coefficient of bmi ≈ 0.07

$$\log \frac{p_B}{1-p_B} - \log \frac{p_A}{1-p_A} = (36.6 - 27.5) \times \hat{\beta} = 9.1 \cdot \hat{\beta} \approx 0.642$$

$$\frac{\frac{p_B}{1-p_B}}{\frac{p_A}{1-p_A}} \approx e^{0.642} \approx 1.9$$

$$se(9.1\hat{\beta}) = 9.1 \cdot se(\hat{\beta})$$

The 95% CI for $9.1 \cdot \hat{\beta}$ would be

```
se.beta<-sqrt(summary(r)$cov.unscaled[7,7])
ci.diff<-9.1*r$coefficient[7]+c(-1,1)*qnorm(0.975)*9.1*se.beta
round(ci.diff, digits=3)
```

```
## [1] 0.154 1.130
```

```
round(exp(ci.diff),digits=3)
```

```
## [1] 1.167 3.094
```

Holding other variables constant, an increase in BMI from the first quartile (27.5) to the third quartile (36.6) is associated with a 0.642 increase in the log odds of testing positive for diabetes (95% CI: 0.154–1.130).

Alternatively, holding other variables constant, an increase in BMI from the first quartile (27.5) to the third quartile (36.6) is associated with a 1.9-fold increase in the odds of testing positive for diabetes (95% CI: 1.167–3.094).

(d)

The estimated coefficient for diastolic blood pressure is -0.00142, suggesting that higher diastolic pressure is associated with a lower probability of testing positive.

```
mean(pima$diastolic[pima$test==1],na.rm=TRUE)
```

```
## [1] 75.32143
```

```
mean(pima$diastolic[pima$test==0],na.rm=TRUE)
```

```
## [1] 70.87734
```

While descriptive statistics show that the test-positive group has a higher mean diastolic pressure (75.32) compared to the test-negative group (70.88), this apparent discrepancy is reconciled by the model's p-value. With a p-value of 0.9, the effect is statistically non-significant, meaning there is insufficient evidence to conclude that

diastolic pressure has a meaningful impact on the test outcome. Thus, the model results do not contradict the descriptive findings.

(e)

Remove observations with missing values before executing step().

```
pima.rm.na<-pima[rowSums(is.na(pima))==0,]  
dim(pima.rm.na)
```

```
## [1] 392  9
```

There are 392 valid observations with complete data across all variables.

```
r.rm.na<-glm(test~.,family="binomial",data=pima.rm.na)  
r.rm.na.f<-step(r.rm.na)
```

```
## Start:  AIC=362.02
## test ~ pregnant + glucose + diastolic + triceps + insulin + bmi +
##      diabetes + age
##
##           Df Deviance    AIC
## - diastolic  1   344.04 360.04
## - insulin    1   344.42 360.42
## - triceps    1   344.45 360.45
## <none>       344.02 362.02
## - pregnant   1   346.24 362.24
## - age        1   347.55 363.55
## - bmi        1   350.89 366.89
## - diabetes   1   351.58 367.58
## - glucose    1   396.95 412.95
##
## Step:  AIC=360.04
## test ~ pregnant + glucose + triceps + insulin + bmi + diabetes +
##      age
##
##           Df Deviance    AIC
## - insulin    1   344.42 358.42
## - triceps    1   344.46 358.46
## <none>       344.04 360.04
## - pregnant   1   346.24 360.24
## - age        1   347.60 361.60
## - bmi        1   351.28 365.28
## - diabetes   1   351.67 365.67
## - glucose    1   397.31 411.31
##
## Step:  AIC=358.42
## test ~ pregnant + glucose + triceps + bmi + diabetes + age
##
##           Df Deviance    AIC
## - triceps    1   344.89 356.89
## <none>       344.42 358.42
## - pregnant   1   346.74 358.74
## - age        1   347.87 359.87
## - bmi        1   351.32 363.32
## - diabetes   1   351.90 363.90
## - glucose    1   411.11 423.11
##
## Step:  AIC=356.89
## test ~ pregnant + glucose + bmi + diabetes + age
##
##           Df Deviance    AIC
## <none>       344.89 356.89
## - pregnant   1   347.23 357.23
## - age        1   348.72 358.72
## - diabetes   1   352.72 362.72
## - bmi        1   360.44 370.44
## - glucose    1   411.85 421.85
```

```
summary(r.rm.na.f)
```

```
##
## Call:
## glm(formula = test ~ pregnant + glucose + bmi + diabetes + age,
##      family = "binomial", data = pima.rm.na)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -9.992080   1.086866  -9.193  < 2e-16 ***
## pregnant     0.083953   0.055031   1.526  0.127117
## glucose      0.036458   0.004978   7.324  2.41e-13 ***
## bmi          0.078139   0.020605   3.792  0.000149 ***
## diabetes     1.150913   0.424242   2.713  0.006670 **
## age          0.034360   0.017810   1.929  0.053692 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 498.10  on 391  degrees of freedom
## Residual deviance: 344.89  on 386  degrees of freedom
## AIC: 356.89
##
## Number of Fisher Scoring iterations: 5
```

Backward variable selection was performed, resulting in the sequential removal of diastolic, insulin, and triceps. The final model obtained is as follows:

$$\log \frac{\hat{p}_i}{1 - \hat{p}_i} = -9.992 + 0.084 \cdot (\text{pregnant}) + 0.036 \cdot (\text{glucose}) + 0.078 \cdot (\text{bmi}) + 1.151 \cdot (\text{diabetes}) + 0.034 \cdot (\text{age})$$

(f) Use the models from (b) and (e), predict the outcome (the probability of testing positive) for a woman with predictor values 1, 99, 64, 22, 76, 27, 0.25, and 25 (same order as in the dataset). Provide a confidence interval for each prediction.

(f)

```
newwoman <- data.frame(
  pregnant = 1,
  glucose = 99,
  diastolic = 64,
  triceps = 22,
  insulin = 76,
  bmi = 27,
  diabetes = 0.25,
  age = 25
)

pred.b <- predict(r,newdata=newwoman,se.fit=TRUE)
eta <- pred.b$fit
se <- pred.b$se.fit

ci.eta <- eta + c(-1,1)*qnorm(0.975)*se

(p.hat <- plogis(eta))
```

```
##          1
## 0.04573331
```

```
(ci.p <- plogis(ci.eta))
```

```
## [1] 0.02502598 0.08213121
```

```
pred.e <- predict(r.rm.na.f,
                  newdata=newwoman,
                  se.fit=TRUE)
eta.e <- pred.e$fit
se.e <- pred.e$se.fit

ci.eta.e <- eta.e + c(-1,1)*qnorm(0.975)*se.e
(p.hat.e <- plogis(eta.e))
```

```
##          1
## 0.04554988
```

```
(ci.p.e <- plogis(ci.eta.e))
```

```
## [1] 0.02500855 0.08155204
```

- Using the model from part (b), the estimated probability of testing positive is 0.0457 with a 95% confidence interval of (0.0250, 0.0821).
- Using the model from part (e), the estimated probability of testing positive is 0.0455 with a 95% confidence interval of (0.0250, 0.0815).

(g)

```
phat <- predict(r.rm.na.f,type="response")

pred <- ifelse(phat>0.5,1,0)

(tab <- table(pred,pima.rm.na$test))
```

```
##
## pred    0    1
##    0 232  55
##    1  30  75
```

```
tab[2,2]/colSums(tab)[2] #sensitivity
```

```
##          1
## 0.5769231
```

```
tab[1,1]/colSums(tab)[1] #specificity
```

```
##          0
## 0.8854962
```

Using a cutoff value of 0.5, the model correctly identifies 57.7% of diabetic subjects (sensitivity) and 88.5% of non-diabetic subjects (specificity).

(h)

```
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
##
## cov, smooth, var
```

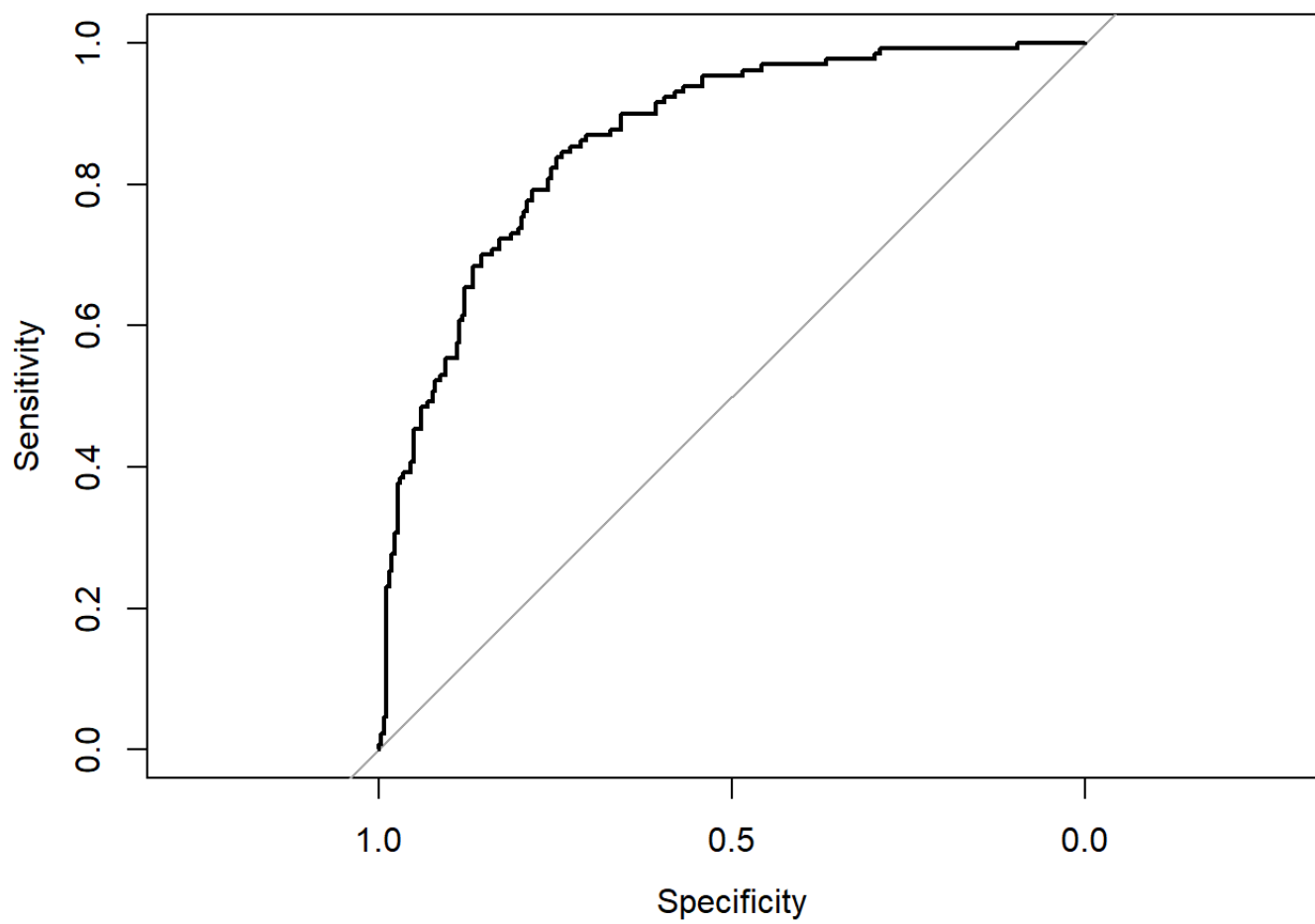
```
phat <- predict(r.rm.na.f,type="response")

roc.obj <- roc(pima.rm.na$test, phat)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
plot(roc.obj)
```

```
auc(roc.obj)
```

```
## Area under the curve: 0.8631
```

AUC = 0.8631.